

AISSAOUI Ismail

AI & Data Science Engineer | PhD Candidate

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PROFILE

AI & Data Science Engineer specialized in **General Purpose AI (GPAI)**, **System 1/2** sequential reasoning, and **Industrial World Models**. Expert in bridging high-fidelity physics with deep learning (Mamba-SSM, PINNs) for real-time edge applications. Proven track record in achieving sub-millisecond inference for safety-critical industrial systems. Eager to advance Smart Manufacturing through GPAI-driven robotics.

EDUCATION

National Higher School of Computer Science (ESI-SBA) 2021–2026
Engineering & Master's Degree in Artificial Intelligence
Master's Thesis: "GPAI-Driven Digital Twin for Industrial Gas Turbines: Bridging System 1/2 for Real-time Prognostics"
Focus: Mamba-SSM, xLSTM, Physics-Informed ML, Edge AI Optimization.

TECHNICAL SKILLS

GPAI & ML: System 1/2, State-Space Models (Mamba), xLSTM, PINNs, GNN, Reinforcement Learning (DQN), XAI.
Robotics & Simulation: ROS, Webots, Robot Kinematics, Navigation, Manipulation, Synthetic Data Gen.
Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, ONNX, MLflow, Airflow.
Edge & Performance: C++ JIT Compilation, CUDA, Docker, Linux, Real-time Stream Processing (InfluxDB).
Full-Stack: FastAPI, Next.js, React, TypeScript, PostgreSQL, Neo4j, WebSockets.

RESEARCH & PROJECT HIGHLIGHTS

Industrial Gas Turbine Digital Twin (Master's Thesis) 2025–2026
Robert Bosch Target: System 1/2, World Models, Edge AI

- Implemented a **System 1/2** architecture for the GE LM2500 turbine: TCNs for rapid perception (System 1) and **Mamba-SSM/xLSTM** for logical trajectory reasoning (System 2).
- Developed a **Generative World Model** for "What-If" counterfactual simulations, predicting physical failures with high-fidelity thermodynamic consistency using **PINNs**.
- Achieved a record-breaking **0.04ms inference latency** via C++ JIT optimization for edge-based control.
- Engineered **XAI heatmaps** using gradient saliency to provide interpretable failure diagnostics.

GNN-Powered Recruitment Ecosystem (Career Connect) 2024–2025
Robert Bosch Target: Multimodal Perception, Graph-based Reasoning

- Modeled the job market as a graph with **PyTorch Geometric**, using GNNs to infer latent structural relationships.
- Integrated **SBERT** and **FAISS** for sub-100ms semantic search across 60,000+ profiles.

Aura — Advanced AI Trading Ecosystem 2025
Robert Bosch Target: Sequential Decision Making, Reinforcement Learning

- Built a magnitude-aware **DQN Reinforcement Learning** agent for automated market participation.
- Developed hybrid LSTM-based temporal models for high-volatility time-series forecasting.

Mini-GPT & Vision Transformer (ViT) Implementations 2025
Robert Bosch Target: Generative AI, Transformer Architectures

- Implemented decoder-only and patch-based transformer architectures from scratch in PyTorch.

LANGUAGES

English: Fluent (C1) | French: Fluent (C1) | Arabic: Native